

3653. XOR After Range Multiplication Queries I

Difficulty: Medium

Link: <https://leetcode.com/problems/xor-after-range-multiplication-queries-i/description/?envType=daily-question>

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Problem:

You are given an integer array `nums` of length `n` and a 2D integer array `queries` of size `q`, where `queries[i] = [li, ri, ki, vi]`.

For each query, you must apply the following operations in order:

- Set `idx = li`.
- While `idx <= ri`:
 - Update: `nums[idx] = (nums[idx] * vi) % (109 + 7)`
 - Set `idx += ki`.

Return the **bitwise XOR** of all elements in `nums` after processing all queries.

Example 1:

Input: `nums = [1,1,1]`, `queries = [[0,2,1,4]]`

Output: 4

Explanation:

- A single query `[0, 2, 1, 4]` multiplies every element from index 0 through index 2 by 4.
- The array changes from `[1, 1, 1]` to `[4, 4, 4]`.
- The XOR of all elements is $4 \oplus 4 \oplus 4 = 4$.

Example 2:

Input: `nums = [2,3,1,5,4]`, `queries = [[1,4,2,3],[0,2,1,2]]`

Output: 31

Explanation:

- The first query `[1, 4, 2, 3]` multiplies the elements at indices 1 and 3 by 3, transforming the array to `[2, 9, 1, 15, 4]`.
- The second query `[0, 2, 1, 2]` multiplies the elements at indices 0, 1, and 2 by 2, resulting in `[4, 18, 2, 15, 4]`.
- Finally, the XOR of all elements is $4 \oplus 18 \oplus 2 \oplus 15 \oplus 4 = 31$.

Constraints:

- $1 \leq n == \text{nums.length} \leq 10^3$
- $1 \leq \text{nums}[i] \leq 10^9$
- $1 \leq q == \text{queries.length} \leq 10^3$
- `queries[i] = [li, ri, ki, vi]`
- $0 \leq l_i \leq r_i < n$
- $1 \leq k_i \leq n$
- $1 \leq v_i \leq 10^5$